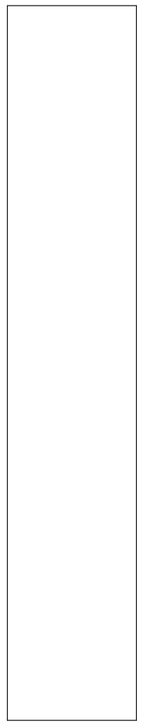
# VAYN shoulder · milling drawing · 16 Ring-1

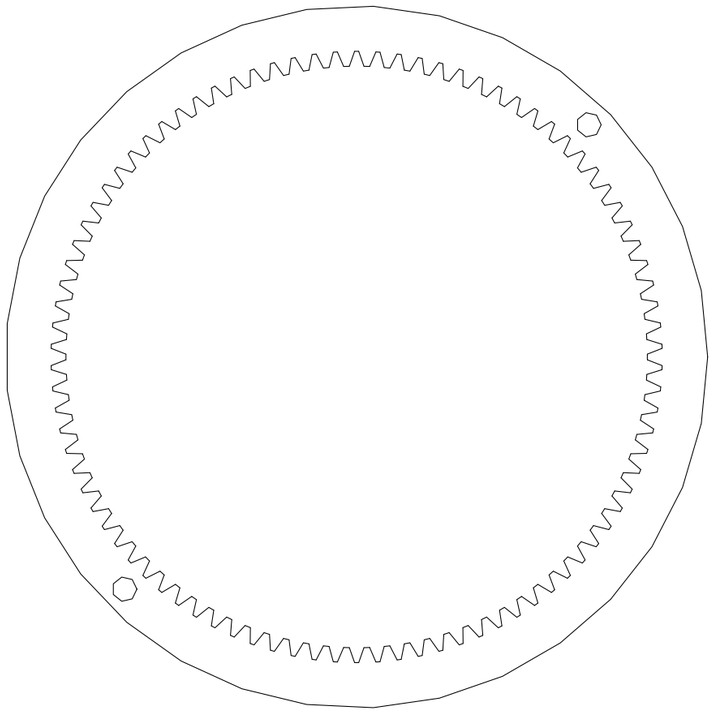


FRONT · Z horizontal, +X up



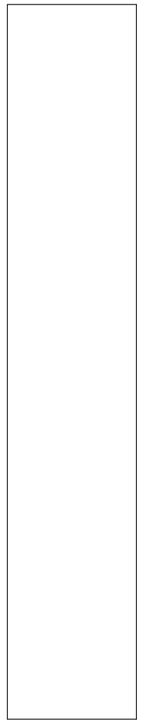
scale 1.37:1 view span 15.4 × 85.0 mm

END · from the arm side (+Z)



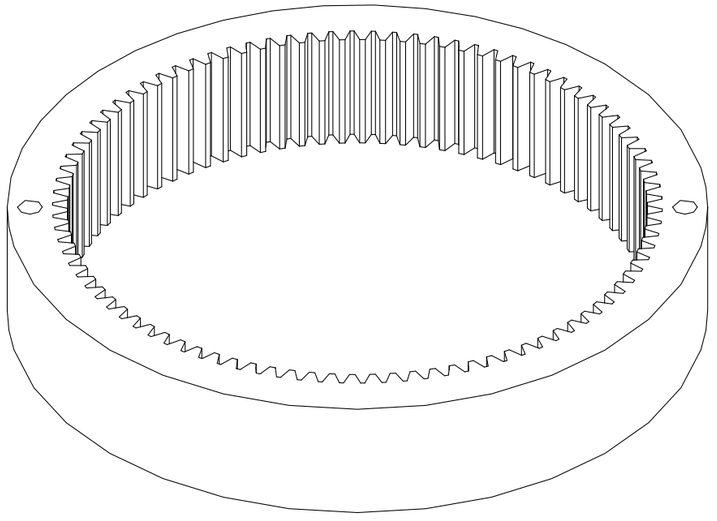
scale 1.35:1 view span 84.8 × 84.9 mm

SIDE · Z horizontal, +Y up



scale 1.37:1 view span 15.4 × 85.0 mm

ISOMETRIC



scale 1.34:1 view span 85.0 × 61.6 mm

## 16

### Ring-1

Quantity in this actuator: 1
Material: 20MnCr5
Group: Gears
Finished size: Ø85 h5 × 15.35 mm
Mass: 190.80 g each
Model bounding box, X × Y × Z: 85.0 × 85.0 × 15.4 mm
Raw stock: Ø105 mm 20MnCr5 bar, annealed, same heat as the other steel
Process: Turn, then wire-EDM the teeth while the steel is still soft
Heat treatment / finish: Carburize the teeth only. Case 0.12–0.18 mm, 58–62 HRC. Mask the outside diameter. No tooth grind. Allow 0.03–0.05 mm growth. Finish Ø85 h5 after heat treatment.
STEP solid: 16 Ring-1 (20MnCr5)

**Controlled features**

- Internal gear, 90 teeth, module 0.8.
- Measure between Ø1.2 mm pins, Mi = 70.813 mm, plus 0.015 / minus 0.
- 2 × Ø3 mm H7 dowel holes, 7 mm deep from the rear face, on Ø79.5 mm.

STEP file is the geometric master. Do not scale this sheet. Sizes not listed here follow ISO 2768-1 class m and ISO 2768-2 class K. Drawing 16_ring_1.pdf